

Gradients with a numerical check

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Slopes and local change

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Parameters and architecture

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Activation functions

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Catch a denominator error for Gradients with a numerical check. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Gradients with a numerical check. Proposed explanation: “The gradient is one scalar”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Check a new example independently

Use $x = 3$, target 2 and $y = wx + b$ at $w = 0$, $b = 1$. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x , 2 times residual].

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Recover a gradient

For the lesson model at $w = 1$ and $b = 1$, calculate prediction, residual, loss and gradient.

My reasoning and evidence

6. W2 Catch a denominator error

A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

My reasoning and evidence

7. W3 Separate a direction from a coordinate

Starting at $[w, b] = [1, 0]$, both coordinates move by plus or minus 0.1 together. Compute the symmetric slope and explain why it is not either partial derivative.

My reasoning and evidence

8. W4 Check a new example independently

Use $x = 3$, target 2 and $y = wx + b$ at $w = 0$, $b = 1$. With step 0.1, calculate both numerical partials. Compare with $[2 \text{ times residual times } x, 2 \text{ times residual}]$.

My reasoning and evidence

9. W5 Recognise a corner

Compute the symmetric difference of ReLU at zero with step 0.1. Explain why this does not prove a derivative exists there.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled gradients with a numerical check evidence sheet to Image-pattern Network Investigation. Show how you can estimate one partial derivative while other parameters stay fixed. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Gradients with a numerical check

Check each coordinate of a gradient while holding the other parameter fixed.

Model $y=2w+b$, target 3. At $[w, b]=[1, 0]$, $y = 2$, residual $y\text{-target}=-1$; loss is residual².

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Gradients with a numerical check

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check w direction

Analytic derivative $2 \times \text{residual} \times 2 = -4$.



2. Check b direction

Analytic derivative $2 \times \text{residual} = -2$.



3. Numerical check

With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

CHANGE ONE THING

Swap the two gradient components in the program output.

Explain the changed result using the matching lesson.

Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.